



**Farm
Sustainability
Assessment**
BY SAI PLATFORM

Outcome Measurement Handbook

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1 Preface

1.1 Why does the FSA have an Outcome Measurement Module?

Farmers and businesses want to show the impact of improving their agricultural practices by measuring on-farm sustainability performance.

There are more and more tools that help measure farm performance. But how to choose the right one? How to apply it in a meaningful way? And how to promote their uptake?

The FSA approach to Outcome Measurement was designed to support farms and businesses in confronting these challenges.

1.2 What does the Outcome Measurement Module consist of so far?

The FSA offers a strong foundation for integrating the use of outcome measurement tools:

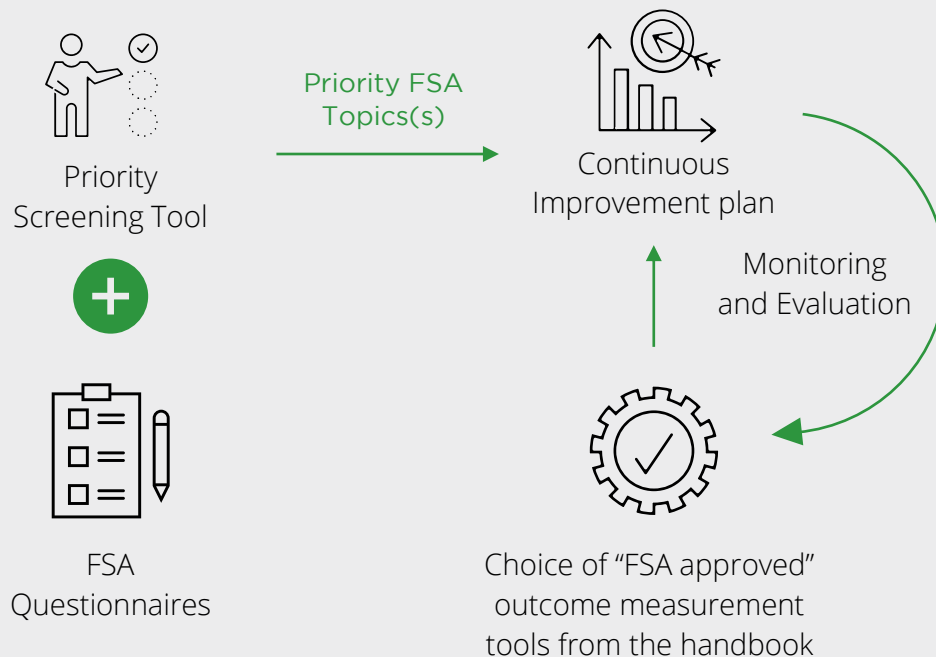
- Proven engagement mechanisms for suppliers and farms
- Verification Framework in place
- Continuous Improvement Module
- Recognized by global food & drink industry, flexible & consistent

The FSA Outcome Measurement Handbook provides guidance on selecting an outcome measurement tool and integrating with your farm sustainability approach. It contains a growing range of available outcome measurement tools that have been assessed by SAI Platform.

The focus of the current version of the handbook includes tools related to outcome measurement for climate/GHGs, soil, biodiversity, and water.

1.3 Where does the Outcome Measurement Handbook fit within FSA implementation?

Outcome Measurement Pathway



- For a given FMG, FSA questionnaires and the Priority Screening Tool will be used to help define the most important sustainability topics to work on, which will be the focus of a Continuous Improvement Plan.
- FMGs are encouraged to use an outcome measurement tool to monitor and support progress of the plan.
- The Outcome Measurement Tools Handbook helps users to decide which tool is most appropriate for their needs:
 - It explains what is available, the ways different tools can be used, as well as reviews and recommends some of the more commonly used tools for each sustainability issue.
 - It should equip FMG coordinators to understand what they want from an outcome measurement tool and differentiate between what is available.

1.4 How will the Outcome Measurement Module develop?

The FSA Outcome Measurement Handbook was initially released in 2020 with the toolset that was part of updates to FSA version 3.0. It is available to anyone seeking to integrate outcome measurement into their continuous improvement plan. This plan is a cornerstone of the FSA, which aims to drive relevant and demonstrable on-farm sustainability performance.

While some farms and businesses will choose to use outcome measurement without further encouragement, others may need additional incentives to start integrating it into their plans. SAI Platform is investigating options for rewarding the use of outcome measurement tools and may more formally incorporate this aspect in the future.

1.5 How can I participate in the development of this module?

We encourage all our members and partners to engage in the development of the outcome measurement module to help drive and demonstrate performance enhancements.

SAI Platform members and their suppliers are welcome to participate, support the governance to help shape this new space or attend workshops or webinars from SAI Platform that are related outcome measurement.

Like the FSA approach to farm sustainability assessment and benchmarking, we intend to build a practical approach for outcome measurement that the industry can align around. This will accelerate the adoption of outcome measurement tools at scale, avoid duplication and measurement fatigue, and build on existing achievements in promoting farm sustainability.

We also welcome engagement from companies or organizations that have developed an outcome measurement tool and that would like to be considered for inclusion in this Handbook. Interested tool owners should contact SAI Platform.

2 SAI Platform Approach to Outcome Measurement

2.1 Why measure?

Outcome measurement is included in SAI Platform's Farm Sustainability Assessment (FSA) program to support farms and Farm Management Groups (FMGs) for their approach to continuous improvement. Outcome measurement tools help to monitor and support progress with the FSA Continuous Improvement Plan¹.

The primary goal of using an outcome measurement tool is to help Farm Management Group (FMG) coordinators and farmers:

1. Understand how farming practices can influence impact measures, such as their greenhouse gas (GHG) emissions, water footprint, soil health, biodiversity, or other quantifiable metrics;
2. Have a rough indication of performance;
3. Provide a tool and mechanisms for engagement between FMG coordinators and farmers; and,
4. Explain the benefits of improvement.

While accuracy of the outcome measurement tool is important, the priority for FSA is to recognize tools that support discussion and shared learning within the FMG. In other words, the right tool used in the right way should start the right conversation – and this dialogue is more important than the result being 100% precise.

In short, outcome measurement can lead to better engagement of all stakeholders in driving and recognizing continuous improvement. It has the potential to create richer conversations between peers, business partners, and stakeholders.

¹ More information on the Continuous Improvement Plan is available in FSA guidance documents.

2.2 Why this handbook?

The main reasons for providing this handbook are to:

- Help FMG coordinators understand at a broad level what they should be looking for from outcome measurement tools in the context of FSA, including pros and cons of different types of tools; and,
- Support FMG coordinators with starting to use a tool that is right for them when considering approaches for their Continuous Improvement Plan.

This handbook has been created with a focus on the process for Farm Management Groups (FMGs). However, elements of the approaches and use of measurement tools are all adaptable also for use by stand-alone farms.

3 General Considerations for Choosing the Right Tool

Measurement tools help quantify and assess the impact of farm practices or change in farming practices on different components of the agroecosystem such as water, biodiversity, soil, and GHG emissions. The use of measurement tools can provide a mechanism for farms and FMGs to help identify impacts associated with different farming practices or technologies. The output from measurement tools can inform where there may be a need for improvement and can provide guidance on the direction of change toward more sustainable practices. They can also be used to help track the success of an improvement plan or project over time. Measurement tools can spark ideas for specific projects that can be initiated to drive improvement. In addition, measurement tools can be used across a group of farmers to benchmark and compare practices to identify where there are opportunities to share best practice and improve the overall performance of the group.

Examples of types of farm-level metrics incorporated in Outcome Measurement Tools:

-  *Biodiversity (e.g., species richness)*
-  *Carbon/GHG Emissions*
-  *Carbon Sequestration*
-  *Energy Use*
-  *Fertilizer Use*
-  *Land Use or Deforestation*
-  *Pesticide Use*
-  *Soil Health or Organic Matter*
-  *Waste or Food Loss*
-  *Water Quality*
-  *Water Use*

**Note:* The above list includes both short term as well as longer term metrics. While a metric like pesticide use would ultimately relate to impacts on biodiversity, human health, water quality, etc., pesticide use metrics could be used as a short-term approach to measure results from practice-based interventions (e.g., adoption of IPM strategy).

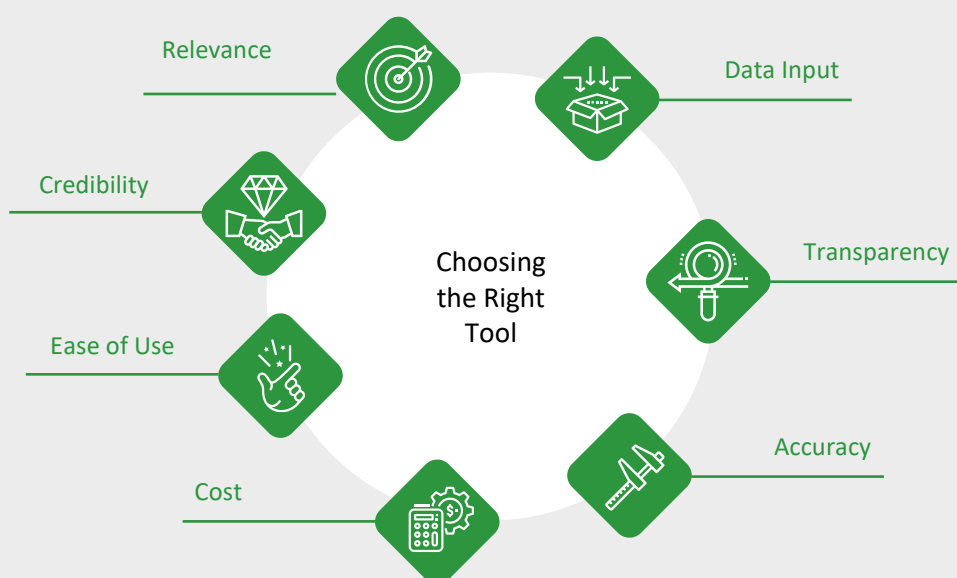
The right tool will depend on the users' purpose for using the tool, for example:

-  Encourage and evaluate continuous improvement or a specific intervention on farms in a supply chain;
-  Produce a 'footprint' to quantify the impact of practices or practice changes (e.g., carbon footprint, water footprint);
-  Evaluate risk mitigation options to reduce negative impacts and model scenarios to identify the most appropriate options to apply in the future; and/or,
-  Provide customer information on the impact of farm practices.

Because of the role of outcome measurement in FSA 3.0 in supporting the Continuous Improvement Plan, the first purpose would be the most relevant for FSA users.

3.1 Factors to consider in selecting an outcome measurement tool

The list below provides a summary of some of the main factors that should be considered when selecting an outcome measurement tool. Emphasis on any specific factor listed below will depend on the purpose for using the tool, as outlined above.



- **Relevance** – Tools must cover the relevant parts of the business's production system and/or supply chain. Many tools are designed for use with annual row crops like corn, soy, and wheat. For farms with other types of crops (e.g., perennial crops, crops grown in a medium other than soil) it is important that the tools ask the appropriate questions that are relevant to the production system.
- **Accuracy** – The tool needs to provide a sufficiently accurate result to support decision making and discussion within the FMG. In some cases, the tools may still be valuable as an approximation of impacts, provided they are used with that understanding.
- **Credibility** – It is important to look for a tool that is based on a robust scientific methodology, has been widely endorsed by experts, is regularly updated, and/or has been accepted by farmers.

- **Transparency** – It is important that measurements tools make information publicly accessible regarding the methodologies used by the tool, responsible parties and governance, and frequency of review and updates.
- **Ease of use** – Measurement tools should be easy to use, and output should be easy to interpret to ensure farmer acceptance. Adequate guidance materials should be provided with the tool in an accessible format.
- **Data Input** – It is important to consider what data is readily available to the farmer to ensure that data entry is manageable and not too difficult to collect, interpret, or upload to the tool.
- **Cost** – To make a tool as accessible as possible to all farmers it is important that it is low cost to the farmer. Economic barriers may be a deterrent to uptake.

There may be additional criteria to consider when choosing the most appropriate measurement tool for the specific needs of a farm or FMG and depending on the crop type and location. It is important in the selection of the tool to consider why the tool was originally developed and for what crops and geographies to ensure that the right tool is chosen for the job. If it is necessary to look at specific crops grown in a particular place, customized local tools, if they are available, could provide the most relevant and specific information. However, if it is necessary to have comparable assessments across a range of geographies a global tool may be more appropriate to allow for cross comparisons.

3.2 Components of outcome measurement

Outcome measurement typically includes three parts: (i) designation of the metric, (ii) direct measurement of outcomes, which may require use of specialized instruments, and (iii) calculation, which may require specialized calculators. These components are defined below:

- **Metric** – a system or standard of measurement, with associated units (e.g., MTCO₂eq of GHG emissions; cubic-meters of water per kg of crop produced; hectares of biodiversity habitat, etc.)
- **Instrument** – the physical device needed for measurement of an outcome (e.g., laboratory equipment needed to measure soil samples for carbon content or water samples for pH, dissolved oxygen, or other features).

● **Calculator** – in the context of farm outcome measurement, an automated web-, excel-, or other software-based calculation tool, developed for quantifying outcomes such as GHG emissions, water, soil health, biodiversity, or other features as they relate to farm operations.

Some outcome measurement approaches do not require a specialized type of physical instrument. For example, counting bird species as part of a biodiversity assessment might only require binoculars. There are also metrics that involve one or more calculations but do not require a special calculator. For example, calculation of water productivity (volume of water per crop yield) is a relatively simple equation and does not require a software program to support the calculation.

The word “tool” in the context of outcome measurement usually refers to either a physical instrument for measuring outcomes or a calculator. The focus in this handbook is primarily related to calculator tools.

4 GHGs and Climate Impact Measurement Tools

4.1 Introduction: Why GHGs and climate impacts matter

Agriculture is responsible for about one fifth (20%) of global greenhouse gas (GHG) emissions². Farm-level activities that are known to have an impact on GHG emissions, include soil management approaches (e.g., application of fertilizers, cover cropping, tillage), irrigation practices, energy use from crop production, harvest, and storage as well as land use changes. International agreements such as the Paris Agreement and the United Nations Sustainable Development Goals have highlighted the role food businesses must play in driving down emissions arising from food production.

² <http://www.fao.org/economic/ess/environment/data/emission-shares/en/>

There are three key GHGs that are associated with agricultural systems; nitrous oxide (mainly from application of nitrogen to soils), methane (mainly produced by ruminant livestock – e.g., cattle and sheep) and carbon dioxide (from the burning of fossil fuels or production of electricity). In recent years there has also been a move towards not just reducing GHG emissions from agriculture, but also improving the capacity for farm and ranch land to sequester and store carbon. In this way, farmers can be viewed as key allies in the fight against climate change and as part of the solution in reducing overall emissions and sequestering carbon.

Outcome measurement tools can support FMGs and farms to better understand the main sources of GHG emissions from their operation and identify opportunities for sequestration.

There are many different approaches for farm-level assessment of GHG emissions including calculators, protocols, guidelines, and models. The focus for FSA purposes is on ‘calculators’ which are defined as automated web-, excel- or other software-based calculation tools developed for quantifying GHG emissions or other outcomes (Denef et al., 2012; *see also section 3.2*). GHG calculators can have a variety of different names, but for simplicity will be referred to as ‘GHG tools’ here.

Using a GHG tool allows the farmer to understand where key emissions sources and sequestration activities exist within current practice. The results from the GHG tool can provide a starting point for discussion between FMG coordinators and farmers on ways to reduce GHG emissions, enhance sequestration, and therefore reduce the climate impact of production. They can also help to visualize and demonstrate potential improvement over time.

4.2 Factors to consider in selecting a GHG tool

In recent decades, growing awareness about GHG emissions from global agriculture and the food production system has created increased pressure on food businesses to understand the scale of GHG emissions within their supply chains back to the farm. This has led to an increase in the development of GHG tools ranging from simple tools requiring only the most basic of farm data and producing only generalized information about GHG emissions, to highly complex tools that provide detailed information on the carbon footprint of the crop or enterprise. There are a range of different tools available to farmers, built for different purposes, using similar but slightly different methodologies and with various assumptions and emission factors used. This means that results differ between tools, even when the same data is entered.

The following sections set out some key factors to consider when selecting a GHG calculator tool, as well as some potentially beneficial additional elements that are available in more complex tools.

4.2.1 Key considerations for GHG tool

As outlined in section (3), the following list includes important points to consider when choosing an outcome measurement tool and as applied to GHG tools. When selecting a tool, it is important to also acknowledge that there are often trade-offs between the considerations. There may be a tool that is specific to your region and relevant to your crop but is not perfect according to other criteria (e.g., ease of use). Ultimately the operation will have to work to find the best solution out of the available options, while considering the points below.

1. **Relevance** – Appropriate for the crop and geography
2. **Accuracy** – Measures key processes that result in emissions from an agricultural production system or for a product
3. **Credibility** – Based on internationally accepted methodology
4. **Transparency** – Assumptions and calculations available for scrutiny
5. **Ease of use** – Provides easily accessible guidance, support materials and is easy to complete
6. **Data Input** – Requires data that is available to farmers and has a simple data entry system
7. **Cost** – Tool-use by farmers is free or inexpensive to access

Relevance – Appropriate for crop and geography

Most tools calculate emissions from annual row crops such as cereals and oilseeds very well, but there is less coverage of perennial crops, protected crops (e.g., greenhouse grown), or substrate grown crops. Tools should provide the relevant types of options for a user's farming system. For example, types of organic amendments available, machinery used, inputs such as plastics or growing media or consideration of refrigerants may be relevant to specific crop groups, especially in the horticulture sector.

Geography is also an important consideration. IPCC allows countries to develop detailed regional emission factors. Regional tools can also more accurately reflect local practice, often making them easier for a user to complete than a generic tool. However, where comparisons are needed across broader geographies a global tool with consistent methodologies and language across all regions is more practical to apply.

Accuracy - Key processes for crop footprint are included

A GHG tool should incorporate all associated processes that contribute to the GHG emissions from crop production in order to properly arrive at an overall number. While there are elements of associated processes that may not be under direct control of the farmer (e.g., irrigation conveyance systems; fertilizer manufacturing), there are generally aspects of all processes that have some degree of control by the farmer meaning that changes in farm practices can have an impact on the emissions coming from each source.

Processes that should be considered by the tool include:

- **Fertilizer emissions from manufacturing** – The manufacturing processes associated with producing fertilizers emit GHGs. The extent of emissions from manufacturing varies depending on the type of fertilizer, where it is produced, and what type of energy is used in the production process. Some tools account for these differences which can provide a useful point of reference and an opportunity to adjust which fertilizers are used on the farm.

- 
Fertilizer emissions from application – Nitrous oxide (N₂O) emissions associated with fertilizer applications are a large part of a farm's carbon footprint, coming directly from soil emissions associated with fertilizer application, and indirectly from volatilization and leaching.
- 
Crop residue management – Crop residues (e.g., the organic material that remains after the harvest of a crop) are a source of nitrogen which can be released as N₂O, but also have the potential to increase soil carbon. GHG emissions arise from breakdown of crop residues. The emissions vary depending on whether the material is incorporated into the soil, burned, or baled so it is important that the tool takes account of how crop residues are handled.
- 
Energy use – Farm activities need energy as farm equipment typically runs on diesel, gas, electricity, or other forms of fuel. The energy required in establishing, managing, harvesting, transporting and storage of the crop should all be captured by the GHG tool. Energy for drying, cooling, heating should also be included when applicable.
- 
Productivity – The tool should take into consideration crop yields and by-products (e.g., straw, wood, low-quality harvest used for animal feed) so that the measurement of GHG emissions can be considered by unit of production and by area (e.g., hectares or acres). In this way, increased productivity can be considered as a mitigation strategy. When emissions can be allocated to by-products, they can reduce emissions from the main product.

Credibility – Based on internationally accepted methodology

The GHG Inventory Guidelines produced by the Intergovernmental Panel on Climate Change (IPCC) is the generally accepted baseline methodology on which most tools are built, with more localized methodologies available in different tiers. However, this can be interpreted in different ways at different levels. There are therefore several standards that have been developed to ensure consistent application of the methodology. The following table provides an overview.

METHODOLOGY	OVERVIEW
Intergovernmental Panel on Climate Change (IPCC) Methodology for GHG Inventories	This is a methodology developed as part of the United Nations Framework Convention on Climate Change (UNFCCC) to support governments in the reporting of their national GHG Inventories. Much of the approach and methodology used in the IPCC guidelines forms the basis of all other GHG emission assessment approaches. IPCC sets out a tier system for emissions factors, such as the examples below. Tier 1 – Global emissions factors Tier 2 – Regional emissions factors Tier 3 – Detailed regional emissions factors
International Organization for Standardization (ISO)	- ISO 14064-1:2018 Part 1: Organization level quantification and reporting of GHG emissions and removals. - ISO 14064-2:2019 Part 2: Project level quantification, monitoring and reporting of activities intended to cause GHG emission reductions or removal enhancements. - ISO 14064-3:2019 Part 3: Specification with guidance for the verification and validation of greenhouse gas statements.
Greenhouse Gas Protocol Product Standard (GHG Protocol)	The GHG Protocol, a collaboration of the World Resources Institute and the World Business Council for Sustainable Development, provides the foundation for sustainable climate strategies and more efficient, resilient, and profitable organizations. The GHG Protocol Product Standard is one of a collection of accounting tools developed by the GHG Protocol to encourage users to understand, quantify, and manage GHG emissions.

Table 4.1. Internationally accepted methodologies for GHG calculation

Transparency – Assumptions and calculations available for scrutiny

All tools must make assumptions, use emission factors, and have methods of calculating the final results based on data inputs from the user. A transparent tool allows the user to review the assumptions to understand how a tool produces a result, and potentially understand why results from different tools differ. Users may rarely use this feature but having it available also shows that the tool owner is confident in the science and approaches used in the tool, by leaving it open to the scrutiny of others.

Ease of use - Provides accessible guidance and is easy to complete

GHG emission assessment is not familiar to all farmers, so it is important that guidance materials are available to explain the data entry requirements, the key elements of the footprint, and how to interpret the results. There are a range of ways this guidance may be offered (e.g., dropdown prompts next to individual questions or sections, links to online manuals or tutorials/webinars, access to online, email or telephone help lines). Materials must be clearly written and concise and available in the relevant language(s).

Tool output needs to be easy to extract by the user, clear in its message even to non-experts, and ideally versatile in what is presented to fit a range of needs. For example, tools should include total emissions, emissions by functional unit, breakdown by the source of emissions and breakdown by type of greenhouse gas. Tools should offer a variety of report options (e.g., excel files, PDF reports, and API integration).

LOCALIZATION:
Availability of multiple languages in both the tool and the guidance enhances ease of use for a range of audiences

Data Input - Has a simple data entry system

Most farmers are familiar with the process of gathering data on their farm operations. Depending on the operation and crop type, many farms may already be using software platforms to capture farm-level data. Using existing data may further streamline data collection, and in some cases, there may be the opportunity to link or integrate data platforms or systems. Data entry should be logical, with dropdowns and help boxes to clarify the types of data needed. Screens should be simple to read with clear and non-technical language that is accessible to all users.

Required data for GHG tools should reflect the type of data that a farmer would have available. Where possible, it may be necessary to make some simplifying assumptions for data elements that a farmer may not know but are needed for the calculations (e.g., volume of residues produced). It is important that the data entry process for calculating GHG emissions requires an appropriate amount of time, considering the potentially limited amount of time that a farmer may have to dedicate to these activities.

The best tools make it clear what data is required at the outset and have well-structured online systems for data entry. The most user-friendly tools provide default values for information that farmers may not have, and some provide industry standards as a guide to the type of values that may be expected. Some tools constrain values that may be entered to prevent unreasonable responses.

STREAMLINED DATA ENTRY:
Prefilled cells and dropdown menus help guide farmers to the type of data needed

Cost – Low and of perceived value

Tools should be low-cost for farmers, allowing at least a limited number of footprints to be done and providing a simple output showing the components of the footprint. Some tools require membership to the organization to gain access and most charge fees for commercial use of their tool. These fees may be based on the size or turnover of the commercial business. Any fees paid by farmers should be perceived as beneficial and valuable to the operation.

4.2.2 Additional considerations when choosing a tool

Beyond the factors listed above, the following list includes additional points that may be beneficial for considering a GHG tool:

- Extra data requirements and tool outputs;
- Capturing changes in land-use;
- Measurement of carbon sequestration;
- Assessment of stored carbon; and/or,
- Advice offered within the tool on mitigation strategies.

These additional measures can give a more complete picture of GHG emissions as well as opportunities for carbon sequestration on the farm. The benefit of using a tool that considers carbon sequestration and carbon storage is to provide a starting point for discussion with farmers on how management practices can have an impact.

Extra data requirements and tool outputs

While a simple carbon footprint can be calculated using the data described in the Data Complexity section above, the inclusion of extra data allows for a more nuanced footprint that captures the impact of additional elements of the production process. Examples of extra data include soil characteristics, plastic used, growing media, refrigerants, and/or capital items. The latter are not normally part of a standard product footprint but can have an impact on emissions.

Additional ways of presenting tool output can encourage more productive conversations between FMG coordinators and farmers and will allow comparisons to be made between farms to highlight the impact of on-farm practices. Helpful additional outputs could include:

- extra detail on sources of emissions, allowing comparisons of data between crops, years, or farms;
- functionality to aggregate data from multiple fields or farms, allowing benchmarking of fields or farms;
- Summary of raw data; and,
- allowing allocation of emissions to by-products of the main crop.

Changes in land-use

Changes in land-use can impact GHG emissions on the farm in different ways. Conversion of land from forest or grassland to crop production or from crop production into grassland is the type of land-use change that has a significant impact on GHG emissions. The IPCC and other GHG accounting standards generally consider the 20 years prior to current production as the appropriate timeframe for monitoring land use change and the resulting change in carbon stock.

Measurement of carbon sequestration

Carbon sequestration is the annual removal of carbon from the atmosphere and the process of storing it in the form of vegetation or soil organic carbon. Land management practices can have an impact on carbon sequestration (e.g., planting trees, hedges, or cover crops; using minimum tillage practices; adding organic amendments to the soil such as crop residues, manure, or compost). Some tools record management practices that contribute to carbon sequestration and include this in the calculation of the carbon balance. Although this can be a useful measurement, the science behind the assessment of sequestered carbon is still developing and the impacts vary depending on soil type, soil conditions, and previous crop management.

Assessment of stored carbon

Stored carbon is the total volume of carbon that is permanently stored in the soils and vegetation present on the farm. For example, both woodlands and soil can store carbon. Most stored carbon is released through changes in land-use (e.g., increased cultivation to a greater depth or wider area, converting from grassland to arable production, soil erosion). Actions that prevent losses to stored carbon could include not cultivating grassland or grass margins, and/or reducing soil erosion (e.g., by maintaining crop cover all year).

Advice offered within tools on mitigation strategies

Guidance materials on next steps and future actions based on the GHG tool output can strengthen the benefits the tool can provide. Knowing where GHG emissions occur within the farming operation is helpful, but for farmers to take the next step to reduce emissions or enhance stored carbon, advice on mitigation options is needed. Some tools provide advice on mitigation or guide the farmer to other sources of information. References of this nature are useful for raising awareness and offering concrete ways for farmers to adjust their practices to drive improvements.

4.3 Example GHG calculator tools

The table below provides a short list of tools that can be used for calculation of farm level GHG emissions. Additional details are provided in Section 8. These are provided as examples, and there are several tools available (e.g., Columb et al., 2013; Deneff et al., 2012; GrainGrowers & Kondinin, 2020; WRI and WBCSD, 2022 - Table III).

Table 4.2. Example calculators available for GHG emissions measurement

Tool Name	Notes
Cool Farm Tool https://coolfarmtool.org/	Comprehensive tool with information on GHG emissions for multiple crops and agriculture systems. Emissions details are provided according to different areas of contributions.
Fieldprint Platform https://calculator.fieldtomarket.org/	GHG emissions are calculated based on data from energy, nitrous oxide emissions from soils, methane (for rice systems), and residue burning. System includes a link to COMET-Planner for scenario planning to review GHG impacts related to on-farm practices.
Puma Platform https://www.plataformapuma.com/	An environmental monitoring platform which supports users to assess crop production cycle impact on carbon footprint and soil carbon balance.

5 Soil

5.1 Introduction

Soil is the dynamic interface between the lithosphere (rock), atmosphere (air), hydrosphere (water), and biosphere (living things). It is the zone in which rocks and organisms, and the air and water that move in and through and around them, interact. Soil is not just the physical parts that make it up, but also the active interactions between its various physical, biological, and chemical parts. Physically, soil is made up of a mixture of materials, including various solids, air, and water in varying proportions.

Managing for soil health is one of the easiest and most effective ways for farmers to increase crop productivity and profitability while improving the environment. Threats to agricultural soils include erosion, salinization, compaction, loss of biodiversity, loss of soil carbon, contamination with toxic substances, and loss to urbanization.

Improvements in soil quality can positively impact soil functioning and support outcomes such as improved soil fertility and nutrient cycling, reduction of plant pathogens, increased water holding capacity, increased storage of organic carbon, transformation of toxic substances, and improved water quality. Soil health should include a measure of soil biology and include repeated measures to detect changes over time.

The FSA includes reference to multiple practices that support soil health, including taking measure to protect soils from disturbance, erosion, compaction, and measures that have been shown to improve soil function such as cover cropping, managing soil organic matter, and protection of soil biodiversity.

5.2 What outcomes are typically measured?

For the purposes of this chapter, outcome measurement for soil health relates to the key measurements that are correlated with optimal soil function for crop growth, such as measurements of plant-available soil nutrients, chemistry, aggregate stability, carbon, compaction, texture, respiration, protein, water holding capacity. Alternatively, qualitative measurements of soil function can be used and scored to evaluate soils.

Example soil health metrics listed in Table 5.1 below are based on suite of metrics used by the Cornell Soil Health Lab (Moebius-Clune et al., 2016), USDA Natural Resources Conservation Service (NRCS, 2015; Stott, 2019; USDA, 2021), and the Soil Health Institute (e.g., Bagnall et al., 2023). The metrics were further reviewed in the context of approaches for sustainable sourcing or regenerative agriculture programs to help refine the list.

Table 5.1. Soil outcome metrics

Metric	Typical Units	Typical Instruments	Notes
Background Metrics Metrics such as pH, texture, and standard nutrient analysis all support interpretation of soil health but may not be an indicator of soil health directly.			
Standard Nutrient Analysis	Macro nutrients N, P and K Micro-nutrients Ca, Mg, S, Fe, Mn, B, Mo, Cu, Zn, Ni, Co, and Si	Lab Analysis	Farmers typically use soil nutrient testing to evaluate nutrient management programs. Not an indicator of soil health on its own.
Soil pH	pH	Lab Analysis	
Soil Texture	% sand % silt % clay	Lab Analysis	Once measured for a site, this value typically does not change. Repeated measures are usually not needed. FAO or governments may provide soil maps of texture.
Soil Health Metrics			
Wet Aggregate Stability	% WSA weight of stable aggregates/ total weight	Lab Analysis	Soils with high wet aggregate stability are more resistant to water and wind erosion and show better overall soil health (e.g., infiltration, root growth, biological activity, etc.).
Total Carbon	% dry weight	Lab Analysis	Measure includes organic carbon as well as soil carbon bound in inorganic carbonate minerals (e.g., calcium carbonate or lime), which is referred to as soil inorganic carbon (SIC).
Soil Organic Carbon (SOC)	% dry weight	Lab Analysis	This metric is used by many soil health standards as an indicator of good soil structure, microbial activity, available water, and available nutrients. SOC is a component of Soil organic matter (SOM), but SOM includes not only the weight of carbon (generally 58%), but also the weight from organically bound water, nitrogen, and other nutrients. SOM is typically estimated from a SOC measurement. Soil organic carbon consists of relatively available organic carbon such as fresh plant residues and more stable organic carbon that is protected in the soil. Soils with a pH below 6.5 are presumed to have no soil inorganic carbon (SIC).

Metric	Typical Units	Typical Instruments	Notes
Permanganate Oxidizable Carbon (POXC) ("Active Carbon")	mg/kg	Lab Analysis	This test measures the portion of soil organic carbon that is easily oxidizable and biologically active, leading to availability for soil microbes.
Soil Respiration (Carbon Mineralization)	CO ₂ released as • lbs/acre/day • kg/ha/day	Lab Analysis	Respiration is a broad and integrative indicator of general microbial activity within soil and can be used for assessment of soil health. Also used as an indicator of total soil microbial biomass. Testing respiration is typically less expensive and more widely available compared to direct measurement of soil microbes.
Predicted Autoclave-Citrate Extractable (ACE) Protein Test	mg protein / gram of soil	Model	Since 2020, Cornell Soil Lab uses their model to predict ACE Protein. Protein content, as organically bound N, influences the ability of the soil to store N, and makes it available by mineralization during the growing season. Soil protein content has also been associated with soil aggregation and thus water storage and movement.
Predicted Available Water Capacity	Grams of H ₂ O per gram of soil	Model	Used by NRCS and others, water capacity serves as an indicator of erosive potential, biodiversity, and the capacity of a soil to store and cycle nutrients. Predicted Available Water Capacity is an indicator of the range of plant available water the soil can store. The lower limit, called the <i>permanent wilting point</i> , is when the soil no longer has plant-available water, the upper limit is <i>field capacity</i> is when the soil can no longer hold more water and water drains via gravity. This measure can also be determined directly using a lab test.

A list of in-field soil metrics is provided in Table 5.2. The list is modified from Moebius-Clune et al (2016). Further information on implementation of in-field approaches can be found in Moebius-Clune et al (2016) and USDA (2021).

Table 5.2. In-field soil outcome metrics

Metric	Typical Units	Typical Instruments	Notes
Earthworms	Poor-medium-good, based on worms per shovelful	Shovel and counts	Earthworms have been identified as an indicator within the scientific literature and counts can be used to track soil health over time. Measure in spring or fall.
Organic Matter Color	Poor-medium-good, based on color of topsoil and subsoil	Visual inspection of color	
Organic Matter Roots/Residue	Poor-medium-good, based on root and residue	Visual inspection or roots and/or residue	
Subsurface Compaction	Poor-medium-good, based on insertion of wire flag	Visual inspection of wire flag during insertion	Best pre-tillage or postharvest and when soil moisture is good.
Soil Compaction	Pressure • kPa • PSI	Penetrometer	
Soil Tilth Mel-lowness Friability	Poor-medium-good, based on texture	Visual inspection of soil properties	
Erosion	Poor-medium-good, based soil behavior after rain	Visual inspection of rills and gullies	Measure after heavy rainfall.
Water Holding Capacity	Poor-medium-good, based on plant stress post rain or irrigation	Visual inspection based on length of time after soil saturation that plants appear stressed.	Water holding capacity is an indicator of erosive potential, biodiversity, and the capacity of a soil to store and cycle nutrients. Measured during growing season.
Drainage, Infiltration	Poor-medium-good, based on pooling post rain or irrigation	Visual inspection of water laying on soil surface	Measure after rainfall or irrigation.
Crop Condition	Poor-medium-good, based on visual cues	Visual inspection of crop color and health	Measure during growing season.
pH	Poor-medium-good, based on how easily pH can be optimized for the crop.	Soil Test Kit or Laboratory	Measure anytime, but at same time of year each time.
Nutrient Holding Capacity	Poor-medium-good, based on repeated soil testing results, fertilizer applied and crop removal of nutrients	Soil Nutrient Testing and crop loss calculations.	Measure over a five-year period, always at same time of year.

5.3 Factors to consider in selecting a tool or measurement approach

This section addresses the main factors to consider in selecting a tool or measurement approach for soil outcomes. The metrics used may depend on the local context of the farm or FMG. In some cases, a farm or FMG may want to start with a broad range of metrics to determine which is most applicable to their context and approach to continuous improvement. Farms or FMG may also leverage relevant organizations to help support the decision to confirm the most appropriate metrics for their site. For example, the Soil Health Institute has recommended three indicators to support scalable assessment of soil health in the context of North American agriculture, including soil organic carbon, carbon mineralization potential (soil respiration), and aggregate stability (Bagnall et al., 2023) as described in Table 5.3 later in this chapter.

Relevance – Appropriate for crop and geography

Farming can impact soil health even when the crop is not grown in soil. Crops grown in media or hydroponics are not typically evaluated for soil health, but erosion and soil contamination are always relevant. Soil-grown crops typically include a nutrient management and soil management plan to manage and amend the soil to match production goals. Soils are heterogeneous and can vary within a single planted area therefore, multiple measurements should be taken that consider underlying geography and past management practices. Generally, soil sampling should be done on a field-by-field basis accounting for specific crop production goals. Governments and research institutions use soil maps to broadly define soils with specific crop suitability.

Accuracy – Key processes are included

Many producers already manage soil nutrient levels and pH using standard soil testing, therefore the constraints on soil health are more frequently found in physical properties of biological health which have either not previously been tested or managed for optimization. Soil health can be difficult to extrapolate from a sample location to a broader geography because of underlying soil heterogeneity. Most producers have already matched nutrient availability and drainage to crop needs but have yet to focus on other soil functions that make the soil more resilient to change while at the same time, ensuring soil

biology is optimized to help meet crop production goals. The accuracy of standardized lab methodologies compared to in-field assessment metrics is still a debate within the academic community. However, most agree that percent soil carbon (SOC) or the estimated value of SOM based on SOC is a valid proxy measurement for tilth and microorganism activity, other measurements such as earthworm counts, color of topsoil to subsoil can also serve as proxy measurements for microorganism activity and drainage. More complex measurements have come in and out of favor. Generally, arable land has sufficient desirable soil properties for crop production, and measurements of soil health are best viewed as directional (increasing, flat or decreasing) to alter management plans so that the soil will be viable for continued agricultural uses. Note that SOC is unlikely to change over a short period of time for a given soil which is the reason behind including multiple variables including wet aggregate stability and biological activity (respiration also known as carbon mineralization, earth worm activity, etc.) as part of soil health outcome measurement.

Credibility – Based on internationally accepted methodology

Because soil is a complex physical and biological system, most academics and governments do not use a single soil variable as an indicator of soil health. There is not an internationally accepted test or measurement. However, aggregate stability, soil organic carbon and soil microbial activity (measured as respiration or other proven proxies) are foundational to soil health. Results from either an experienced soil scientist's evaluation of soil texture using the visual inspection method to determine soil tilth, mellowness and friability, or a lab test of soil wet aggregate stability are repeatable and accurate but cannot be compared directly. Measuring differences over time using similar methods is critical to acceptance.

Transparency – Assumptions and calculations available for scrutiny

Soil testing labs that make their methodology publicly available are preferred. In-field techniques should be from published methodology from reliable sources where the methods are documented or published in a peer reviewed journal. For example, soil health sampling packages from the Soil Health Institute and Cornell University include standard operating procedures (SOPs) of their methods. SOPs should

include how to take soil samples in the field, how to store and transport samples, and the lab protocols used.

Ease of use – Provides accessible guidance and is easy to implement

Sending soil samples to a lab for testing can be expensive and usually requires special equipment (e.g., soil probe, sterile sampling equipment, dry ice). In-field sampling requires expertise to reliably evaluate soil over time. In-field techniques may be soil-type dependent and practitioners in one locality might need training to evaluate soils from a different location. Farmers using crop advisors or nutrient service providers typically take soil samples at an interval of 1-3 years. Many service providers offer soil sampling. Adding specific tests, such as total organic carbon, soil respiration, and wet aggregate stability and increasing sampling frequency will depend on cost and availability of local tests or practitioners.

Data Input – Has a simple data entry system

Soil lab testing routinely includes results as a report and sometimes outputs are available digitally. Soil health evaluations require repeated measures to detect changes over time. There is no standardized methodology to compare values over time. Storing values in a database, spreadsheet or other computer-file based system that includes some visualization capability can make the results more accessible.

Cost – Low and of perceived value

Farmers using fertilizer usually already have recognized the value of soil testing as part of an overall nutrient management plan. The added expense of the additional physical and biological tests may be justified through management activity that results in increased soil function, such as reduced nutrient loss, increased water holding capacity, reduced erosion, etc.

5.3.1 Additional considerations when choosing a tool or measurement approach

Sending soil samples to a lab using a standard protocol is the most reliable approach, but also the most expensive. Using in-field measurement is also possible, but many such measurements can be subjective. Evaluating cost, access to a reliable soil sampling lab and available local expertise will influence the measurement approach.

Timing. While the timing of measurement of soil fertility for the purposes of crop management (e.g., to guide nutrient recommendations) may be clear based on when crop nutrients are needed, the timing of measurements for soil health may be less obvious. There can be seasonal variability in some metrics of soil health, therefore it is important to be consistent in the timing of measurement of soil health for comparisons between farms or over time. Practical considerations (e.g., team availability; busy time of year for farm activities, etc.) may determine the best time for taking soil health measurements, which should be considered in developing a protocol to ensure that soil health tests can occur at a consistent time for the FMG project.

Aggregation of Metrics. Like other types of outcome measurements (e.g., biodiversity) there are examples of soil health metrics that represent an aggregation of component metrics. For example, Cornell combines many of the soil metrics listed in Table 5.1 to render an “Overall Quality Score” which can be compared for fields in the same geography but under different management practices. The advantage of aggregate metrics is that they provide a single number that can be compared between sites or over time. However, a potential disadvantage is the difficulty in interpreting differences in the number if there are multiple factors that can influence it.

TIMING OF SOIL HEALTH MEASUREMENT:
To ensure consistency of measurement, make sure that soil health measurements are taken at a similar time each crop year and at a time that works best for the operation.

5.4 Example metrics and calculators for soil

Table 5.3 below provides examples of available metrics and approach to soil measurement in each. In most cases, available approaches for soil outcome measurement rely on collection of metrics rather than a calculator tool. All examples below reflect groups of metrics, except for one calculator (Fieldprint Platform).

Table 5.3. Metrics/calculators for soil health outcome measurement

Tool Name	Relevant Metric(s)	Notes
Comprehensive Assessment of Soil Health – Lab Sampled (Cornell)	- See quantitative metrics in Table 5.1. Main examples include: - Soil Organic Carbon - Wet Aggregate Stability - Soil Respiration - And others	Cornell approach represents a collection of metrics rather than calculator tool.
Comprehensive In-field Soil Health Assessment Metrics (Cornell)	- See quantitative metrics in Table 5.2. Main examples include: - Earthworms - Organic Matter - Soil Tilth - Water Holding Capacity - And others	Cornell approach represents a collection of metrics rather than calculator tool.
Soil Health Institute	- See quantitative metrics in Table 5.1. SHI has identified three indicators as broadly applicable in North America: - Soil Organic Carbon - Wet Aggregate stability - Soil Respiration	SHI approach represents a collection of metrics rather than calculator tool.
Fieldprint Platform	- Erosion (modeled loss potential) - Erosion potential from wind - Erosion potential from water - Soil loss (weight) / area/ year	Fieldprint uses soil conservation metric in tons of soil lost (T) per unit of land area (acre) per year. Modeled from NRCS (WEPP, WEPS), with soil erosion from wind and water in T / ac / yr.

6 Biodiversity

6.1 Introduction

Agricultural production represents about 35% of global land area (Ritchie and Roser, 2013). Roughly one third of agricultural land is used for crop production³ (FAO, 2022a). Recent estimates have indicated that land-use conversion for agriculture is responsible for close to 90% of the world's deforestation (FAO, 2022b) and agriculture has been cited as the biggest driver of habitat loss (Williams et al., 2020). At the same time, overall improvements in agricultural productivity alongside a growing population have resulted in a trend of decreasing cropland area per capita (FAO, 2022a). In addition to impacts due to land conversion and habitat fragmentation, agriculture can also influence surrounding ecosystems because of water use in irrigation, water runoff from farm fields, pesticide drift, and other aspects related to farm management. Recent global initiatives (e.g., UN Biodiversity COP 15) and trends (e.g., nature-positive approaches (e.g., zu Ermgassen et al., 2022)) have supported an increased focus on biodiversity-related commitments by the public and private sector, including the creation of goals for habitat conservation and promotion of biodiversity in agricultural systems through species conservation.

The FSA includes multiple practices that support biodiversity and habitat, including responsible management of inputs like fertilizers and pesticides, conservation of primary forest and other ecosystem types, land restoration, and other practices.

For the purpose of understanding biodiversity outcome measurement in agriculture, it is important to recognize that agricultural working lands represent anthropogenically modified ecosystems. Unless agricultural lands are fully restored to their original natural ecosystem type, it is generally difficult if not impossible for farming lands to achieve equal ecosystem functioning with equivalent biodiversity attributes, such as type and number of species, as compared to their pre-agricultural state. However, there is an opportunity for agricultural working lands to be managed in a way to provide *similar* functioning

³ As of 2020, total global agricultural land area was 4.74 billion hectares, of which 1.56 billion hectares were cropland and 3.18 billion hectares were used for permanent meadows and pastures (FAO, 2022a).

as the prior natural state (e.g., Lind et al., 2019). In other words, farming will necessarily have an impact on the ecosystem and related species, but a key objective for sustainable agricultural systems is to minimize negative impacts while maximizing positive impacts from agricultural operations on biodiversity outcomes.

6.2 What outcomes are typically measured?

Biodiversity outcomes typically measured on agricultural operations can be grouped into four general categories: *species diversity*, *habitat or land*, *ecosystem services*, and *farm management*. These are shown in Table 6. 1.

Table 6.1. General categories of biodiversity metrics

	Species Diversity		Ecosystem Services
	Habitat/Land		Farm Management

Species Diversity. Species diversity can be measured in various ways, including the total number of different species (species richness), relative abundance of species (species evenness), as well as metrics that incorporate evolutionary relatedness (phylogenetic diversity). Measurement of species diversity may also include focus on key indicator species, rather than all species, as a way to evaluate and track ecosystem health.

To evaluate species diversity in the context of a farm or farm group, the regional scope of evaluation could be limited to the boundaries of the land owned or managed by the farm and/or could include adjacent land areas outside of the farm. Metrics that are focused within the farm boundaries directly can include diversity within productive areas of the farm (e.g., crop rotation, varieties/species of crops grown, cover crops) or diversity in non-productive areas that are owned or managed by the operation (e.g., field margins, hedgerows, fallowed land, riparian buffer strips). Impacts of the farm may also be assessed through evaluation of species, habitat, or ecosystem services on areas that are adjacent to the farm but under separate management or ownership.

A summary of the types of species diversity that could be included in biodiversity assessment are in Table 6.2 below. The table provides an overview according to whether species are terrestrial (above- or below-ground), and/or aquatic (associated with on-farm or adjacent waterbodies).

Table 6.2. Examples of different types of species diversity related to agricultural production systems

		Terrestrial		Aquatic
		Above-Ground	Below-Ground	
Farm Productive Areas	Crop	Intercropping or crop rotation	Soil biodiversity, including bacteria, fungi, insects, earthworms, other invertebrates.	NA
	Non-Crop	Cover crops; small habitat areas (e.g., hedgerows, riparian buffer)	Soil biodiversity, including bacteria, fungi, insects, earthworms, other invertebrates.	Applicable if water bodies exist on the farm. Includes invertebrates, algae, and other aquatic organisms.
Outside of the Farm		Flora and fauna of land areas adjacent to the farm	Soil biodiversity in areas adjacent to the farm	Flora and fauna of water bodies adjacent to the farm

Habitat/Land. Measurements of habitat typically focus on the extent of land area that is conserved or restored, number of habitat types, habitat quality, and connectivity. Generally, measurement of land area is the most feasible of biodiversity metrics and one of the most important biodiversity metrics due to the fact that habitat loss is the key driver of species extinction (e.g., Williams et al., 2020). Habitat quality can include various features, such as biological, physical, and chemical properties. Habitat connectivity is a feature on a broader regional scale and describes how patches of habitat are connected in a landscape. Habitat connectivity metrics are generally more relevant for Farm Management Groups (FMG) rather than a stand-alone farm, though can apply in some cases for very large farms or plantations.

Ecosystem Services. Ecosystem services represent any direct or indirect benefit that nature provides to society. In the context of agriculture, ecosystem services can include many different attributes, such as carbon sequestration, maintenance of air and water quality, support for pollination and seed dispersal, buffering of extreme weather events, nutrient cycling and more (IPBES, 2019). In this context, agricultural productive land can provide positive ecosystem

services (e.g., carbon sequestration) even if it is highly modified from the natural state. The challenge, however, is that ecosystem services may be difficult to measure. Ecosystem service metrics may also overlap with other outcome metrics addressed in this Handbook (e.g., water, soil).

Farm Management. Metrics in this category typically represent an aggregate of information about a farm, including farm practices and other attributes. As such, biodiversity metrics that are in this category may not reflect an outcome but rather actions or potential that enable or support biodiversity outcomes.

Examples of metrics used for biodiversity outcome measurement are provided in Table 6.3. With some exceptions (e.g., habitat connectivity), the examples are mostly focused on outcome metrics applicable to a farm or farm group, rather than those applicable at a broader landscape or regional level.

Table 6.3. Biodiversity outcome metrics

Metric	Typical Units	Typical Instruments	Notes
Species			
Species Count (i.e., species richness)	Number of species, by type (e.g., flora, fauna, etc.)	Ecological assessment; below-ground measures typically require genetic testing/DNA sequencing to assess microbial biodiversity	Approaches may distinguish between native and introduced species, including those that are considered invasive. An assessment may focus on indicator species rather than carry out a full biodiversity inventory (e.g., Billeter et al., 2007). The species diversity metric may also consider both crop and non-crop species.
Other species metrics	Varies	Varies	There are a variety of metrics to evaluate species diversity beyond the species count. Species evenness is similar to a count of species, but also accounts for relative abundance of different species. Biomass has been used as a proxy for below-ground species diversity estimates (e.g., fungal biomass). There are a variety of metrics that could be considered here (e.g., phylogenetic species diversity accounts for evolutionary history; Shannon Index, etc.). Jennings et al. (2020) provide some additional examples of species metrics.

Habitat			
Total Area	Land that is conserved or restored habitat, or providing positive ecosystem services (area or %)	Land survey, remote sensing/satellite imagery	Metric could include non-cropped areas of the farm such as field margins or buffer strips, conserved areas within the owned/managed farm, or adjacent areas.
Connectivity	Score of connectedness or fragmentation of habitats, multiple metrics have been proposed (Keeley et al., 2021)	Typically requires mapping of habitat areas	Metric is usually applied at a landscape level.
Quality	Biological, physical, or chemical characteristics of the habitat	Survey of indicator species; physical or chemical characteristics of soil or water (e.g., concentration of pollutants), etc.	Habitat quality metrics may be varied and contingent on original ecosystem type as well as what aspects of the habitat may be relevant.
Ecosystem Services			
Various types of outcome measurement can be included under the category of ecosystem services which generally reflects the functional output from biodiversity. Examples may include nutrient cycling (e.g., nitrogen), carbon sequestration, buffering against extreme events, pollination support, development of soils and others (examples in IPBES, 2019). Given the variability of types of ecosystem services, these are not included in further detail in this document but may be a point for a farm or FMG to review for applicability to their system and operation.			
Farm Management			
Multiple aggregate metrics related to biodiversity have been developed with examples provided here. As noted above, metrics that include reference to farm management practices typically incorporate consideration of actions that support biodiversity outcomes, rather than the outcomes directly.			
Cool Farm Tool Biodiversity Metric	Multiple scores as % for each biodiversity category.	Assessed with the Cool Farm Tool.	Metric quantifies how well farm management supports biodiversity.
Habitat Potential Index (HPI)	Score between 0 and 100, with 100 being most positive.	Assessed with the Fieldprint Platform.	Metric based on attributes of the farm, management practices, and changes in land use.
Healthy Farm Index (HFI)	Value between 0 and 1 reflecting aggregate of multiple biodiversity and ecosystem service metrics.	Can be calculated as described by Quinn et al. (2013) or with associated calculator.	Metric that includes multiple indicators on biodiversity and ecosystem services, both practice-based as well as metrics related to species diversity.

6.3 Factors to consider in selecting a tool or measurement approach

This section addresses the main factors to consider in selecting a tool or measurement approach for biodiversity outcomes.

Relevance – Appropriate for the crop and geography

Biodiversity outcome metrics, such as habitat area and species diversity, are generally relevant to all crop types and geographies. Details of the specific measurement approach could be impacted by the natural ecosystem type of the region where the farm is located (e.g., forest, grassland, wetland, etc.) or if there are both associated terrestrial and aquatic environments.

Accuracy – Key processes are included

There are several ways of measuring species diversity including whether the farm or adjacent areas are included, evaluation includes terrestrial or aquatic, above- or below-ground species, or is focused on specific indicator species. The measurement approach should be researched to ensure that all relevant impacts are assessed, while also recognizing the practical realities of time and budget. Selection of indicator species may be particularly important to confirm since research has indicated that use of indicator species (e.g., individuals or species of a particular group, such as birds) to survey overall biodiversity may have limitations as a helpful predictor for all species types (Billeter et al., 2008).

Credibility – Based on internationally accepted methodology

Measurement of species diversity will require confirmation of methodological approach, including proper sampling or lab techniques, if applicable (e.g., genetic sequencing of soil microorganisms). While measurement of overall conserved or restored land area may have more standardized approaches and require fewer assumptions to implement (e.g., using GIS software), other metrics (e.g., habitat connectivity, aggregate biodiversity index) may require review and confirmation that the methodological approach is justified and based on credible sources.

Transparency – Assumptions and calculations available for scrutiny

The calculation of metrics like species diversity (e.g., species richness) or habitat area is generally standardized, or the calculation is simple enough that there may not be a need to provide all details of the methodological approach (e.g., if the metric is total area of land or number of species). When using aggregate metrics (e.g., Healthy Farm

Index; Habitat Potential Index), results should be accompanied by the description of calculation approach.

Ease of use – Provides accessible guidance and is easy to implement

While metrics on number of species are easy to interpret, they may be harder to measure. For example, biodiversity surveys require expertise with the flora and fauna of an area, and in some cases can only be carried out by a professional biologist. Aggregate metrics related to species, habitat or other features that require multiple steps to calculate should have sufficient documentation to understand and implement.

Data Input – Has a simple data entry system

Other than measures of habitat area, biodiversity outcomes related to species diversity are likely not regularly recorded by a farm operation. There are examples of biodiversity assessment tools that rely on more easily obtainable information to understand potential impacts of the farm on biodiversity (e.g., practices related to pest management). Therefore, there can be tradeoffs between input data that may be more readily available for the farm (e.g., farm practice data) which generally would provide a proxy for biodiversity metrics, and information such as number of native species, which is more difficult to gather but represents a more direct measure of biodiversity outcomes.

Cost – Low and of perceived value

The cost for inventories of species diversity can be quite high, since above-ground biodiversity assessment may require professional biologists to identify the various plant, animal, or other types of species on or adjacent to the farm. Below-ground biodiversity assessment may also require significant costs due to the need to employ technologies like DNA sequencing to identify soil microbial diversity. Furthermore, to be most valuable, measurement of biodiversity should occur at regular intervals and therefore the budget should ideally be available to support regular monitoring. Therefore, a full species assessment may not always be financially feasible for a farm and alternatives such as measurement of total land area (e.g., hectares of conserved land) or practice-based proxies may be more feasible. Examples of practice-based metrics in use by sustainability programs include the Cool Farm

Tool Biodiversity Metric, the Habitat Potential Index (HPI) by Field To Market, and the Habitat and Biodiversity Metric from Stewardship Index for Specialty Crops.

6.3.1 Additional considerations when choosing a tool or measurement approach

Aggregation of Metrics. Outcome measurement can include metrics that represent an aggregation of multiple indicators into a single number. Examples of aggregate biodiversity metrics include the Healthy Farm Index (Quinn et al., 2013) which incorporates various elements of species diversity and ecosystem functioning or the Habitat Potential Index (HPI) which is based on land type, land use changes, and management practices. While a single, aggregate metric like the Healthy Farm Index or Habitat Potential Index can be easier to report and track, aggregate metrics may be more difficult to interpret and potentially more complicated in relating the measure to actions or practices that could improve the outcome.

Landscape Approaches. As with other outcome measurements like water, an important distinction in biodiversity outcome measurement relates to whether the measurement is on-farm or at a broader, landscape level. Some biodiversity outcome metrics (e.g., habitat connectivity) or calculators (e.g., Global Forest Watch) only apply at the level of the landscape. Furthermore, cost and time-intensive measurement of biodiversity outcomes, such as a full species inventory of the farm and surrounding areas may also only apply at a broader regional level. Some landscape approaches rely on remote sensing and may also address other outcome areas. For example, NDVI (Normalized Difference Vegetation Index) can be used to assess vegetation cover through remote sensing and could potentially also be used as a proxy for soil health.

6.4 Example calculators for biodiversity metrics

Table 6.4 provides a summary of examples of available calculators and approaches to biodiversity measurement. Most farm-level tools provide information on metrics that incorporate data on farm management and related practices. General information on some of the tools is provided in a separate section.

Table 6.4. Calculators available for biodiversity outcome measurement

Tool Name	Relevant Metric(s)	Notes
Cool Farm Tool https://coolfarmtool.org/	Biodiversity Assessment	Practice-based metric that provides results for productive and non-productive areas of the farm, distinguished according to small habitats (e.g., hedgerows, field edges) or large habitats (areas of natural habitat larger than 1 hectare). Results include information by species group (e.g., woodland flora, woodland birds, etc.).
Fieldprint Platform https://calculator.fieldtomarket.org/	Biodiversity Metric – Habitat Potential Index (HPI)	Defined by Field To Market as a measure of the potential capacity of the field to support diverse ecosystems, calculated based on type of land, ecological value, and land management (FTM 2022).
Stewardship Index for Specialty Crops (SISC) https://www.stewardshipindex.org/	Habitat and Biodiversity	Practice-based metric that calculates a score for cropped and non-cropped areas of the farm. Metric calculation is available as a document or as software tool.
Biodiversity Performance Tool (BPTi) https://www.biodiversity-performance.org/	Multiple indicators on biodiversity performance	Evaluates the potential for biodiversity at farm level using information on indicators related to farm environment, habitats, and farm practices. Indicators are rated with a traffic light system (red/orange/yellow/green). Applicable to farms in European countries.
Healthy Farm Index (HFI) https://johnquinniv.wixsite.com/agroecology/healthy-farm-index	Healthy Farm Index, and associated metrics	Aggregate index that incorporates practice-based and quantitative metrics (e.g., number of bird species), requiring input of current state, targets, and relative weight/importance of component metrics. The USDA National Organic Program Biodiversity Guidance Compliance Tool is similar to the Healthy Farm Index (HFI). The Biodiversity Compliance Tool was created to support certified organic farms and is based on multiple indicators for biodiversity, including species diversity, habitat area, and practices (e.g., tillage). https://johnquinniv.shinyapps.io/biodiversity_calculator_v06/

In addition to the calculators listed in Table 6.4, there are a variety of tools focused on risk management and decision-making that provide information from datasets on biodiversity attributes generally at a larger landscape or regional level. Some examples are listed below:

- *Global Forest Watch*⁴ is a comprehensive online platform that provides a mapping tool using satellite remote sensing data to monitor forest change and identify trends over time.
- *Integrated Biodiversity Assessment Tool (IBAT)*⁵ links datasets about threatened or endangered species, protected areas, and key biodiversity areas to map locations to provide reports and information to help inform biodiversity risk screening.
- *InVEST (Integrated Valuation of Environmental Services and Tradeoffs)*⁶ is a suite of open-source software tools developed by the Natural Capital Project to help quantify and map the benefits and tradeoffs of natural resource management on ecosystem services.
- *World Wildlife Fund (WWF) Biodiversity Risk Filter*⁷ is a corporate and portfolio-level screening tool to help companies and investors to prioritize action on what and where it matters the most to address biodiversity risks.

6.5 Summary

Calculation approaches for biodiversity measurement may be relatively simple but require a high level of expertise. For example, determination of species diversity is the approach that directly evaluates biodiversity outcomes but requires expertise and potentially the support from a trained biologist. Because of the challenges with obtaining species diversity estimates and the need to account for what is under a farm's control, many tools that evaluate farm-level biodiversity metrics use a proxy of farm management practices as a component of measurement. Landscape level measurement beyond the borders of the farm or farm group may often be relevant or the most appropriate scale to measure habitat or land area through mapping and remote sensing. Recently, options to monitor habitat and land area metrics within agricultural supply chains have been expanding due to need to address regulatory requirements (e.g., European Union Deforestation Regulation) or voluntary commitments related to deforestation (e.g., Science Based Targets Network, SBTN).

⁴ <https://www.globalforestwatch.org/>

⁵ <https://www.ibat-alliance.org/>

⁶ <https://naturalcapitalproject.stanford.edu/software/invest>

⁷ <https://riskfilter.org/biodiversity/home>

7 Water

7.1 Introduction

Agriculture is responsible for about 70% of global freshwater withdrawals (FAO, 2022c) with water consumption expected to increase due to a growing population and associated demands for increased food supply. At the same time, water pollution and water quality are impacted by farm operations through agricultural runoff and other factors such as use of pesticide and fertilizer inputs. Farming operations, therefore, have a critically important role in promoting sustainable water management, ensuring optimal water use, and preserving water quality while meeting global needs for increasing food production. Farm practices that support sustainable water management are also closely connected with other environmental outcomes such as aquatic and terrestrial biodiversity, soil health, and GHG emissions.

The FSA includes reference to multiple practices that support sustainable water management through use of efficient irrigation systems, responsible management of inputs such as nutrients and pesticides, implementation of techniques to minimize runoff (e.g., buffer zones) and many others.

7.2 What outcomes are typically measured?

For the purposes of this chapter, outcome measurement for agricultural water relates to the amount or *quantity* of water used on a farm - primarily applicable to irrigated farms - and the *quality* of water in runoff and associated surface water and groundwater resources. There are a variety of approaches that can be used to measure water quantity and quality outcomes. Typically, the complexity in measurement of water-related outcomes is connected to the type of instruments or laboratory tests required, such as flow meters installed on a farm to measure amount of irrigated water, or laboratory tests that confirm physical or chemical quality characteristics of adjacent water bodies. An additional important consideration for measuring water outcomes is whether to focus on only the farm or farm group directly, or to incorporate outcome measures at the level of the broader watershed or irrigation district (if applicable).

Typical farm-level water outcome measurements include:

- **Water Quantity.** While total water used by the crop from both blue, green, or other sources can be measured or estimated, the typical focus of this measurement is on the blue or irrigation water component. Water quantity can also be assessed through the determination of water balance, considering water in- and out-flows as well as storage.
- **Water Quality.** Water quality can be measured in terms of physical, chemical, and biological attributes. Examples of water quality measurement include pH, Total Phosphorus (TP), Total Nitrogen (TN), Total Suspended Solids (TSS) and others. While methods exist to measure the quality of water runoff (e.g., edge of field monitoring), measures of water quality are usually applied to water bodies adjacent to the farm, such as streams, rivers, ponds, or lakes.

There are additional outcome metrics that relate to behavior of water in soil (e.g., water holding capacity, water infiltration) that are covered elsewhere in the handbook. More details on outcome measures related to water and typical metrics are shown in Table 7.1.

Table 7.1. Water outcome metrics

Metric	Typical Units	Typical Instruments	Notes
Water Quantity			
Applied Water Use Efficiency <i>or</i> Water Footprint <i>“drop per crop”</i>	Applied Water (e.g., gallons, acre-inches, cubic meters) per unit of crop yield (e.g., kilograms, pounds, tons). In some cases, the metric may be expressed as water used per unit area (acres or hectares) rather than yield.	Device (e.g., flow meter) needed to measure applied water; measure of crop yield also required	This metric can usually refer to blue water, but footprint metric can also include green or grey water contributions. Estimated values for water applied can be made using water application rate and duration. The footprint measure is used here to indicate footprint on the farm only, but some approaches measure the total water footprint of a product including production and processing steps. The general term “water-use efficiency” refers to the amount of carbon (either total biomass or yield) produced per unit of water (Hatfield and Dold, 2019).

Water Productivity <i>"crop per drop"</i>	Unit of crop yield (e.g., kilograms, pounds, tons) per applied water volume (e.g., gallons, acre-inches, cubic meters)	Device (e.g., flow meter) needed to measure applied water; measure of crop yield also required	This metric usually refers to blue water withdrawals – both ground-water and surface water – and applies only to irrigated farms. Estimated values for water applied can be made using water application rate and duration.
Irrigation Efficiency	Applied water divided by total water demand of the crop	Device such as flow meter needed to measure applied water; crop water demand requires measurement of reference evapotranspiration (ET) adjusted to the crop coefficient (ETc) accounting for rainfall.	Applicable only to irrigated farms. Estimated values for water applied can be made using water application rate and duration. Regional and historic ET may be available.
Water Balance	Calculated using the formula: water outflow = water inflow + change in storage	Usually calculated on a one-year time scale. Inflow includes incoming water from rain or water supply (e.g., surface water), outflow can include wastewater, run-off, or evaporation.	Metric can apply to both farm and overall watershed. Metric is referenced in the Alliance for Water Stewardship Standard (AWS, 2019).
Water Quality			
Physical	Examples of physical characteristics of water quality: • Temperature • Turbidity - Total Suspended Solids (TSS) • Salinity	Water sampling and associated laboratory tests required.	This metric typically applies to characteristics of surrounding water bodies, though can be applied to edge-of-field monitoring.
Chemical	Examples of chemical characteristics of water quality: • pH • Dissolved oxygen • Biochemical oxygen demand (BOD) • Total Phosphorus (TP) • Total Nitrogen (TN) • Concentration of heavy metals or other pollutants (e.g., pesticides)	Water sampling and associated laboratory tests required.	This metric typically applies to characteristics of surrounding water bodies, though can be applied to edge-of-field monitoring.
Biological	Biological characteristics of water quality typically include quantification of associated aquatic organisms (e.g., algae, aquatic invertebrates).	Water sampling and associated laboratory tests required; may be achieved through edge of field monitoring.	This metric typically applies to characteristics of surrounding water bodies, though can be applied to edge-of-field monitoring.

7.3 Factors to consider in selecting a tool or measurement approach

Relevance – Appropriate for the crop and geography

Measurement of water quantity and quality can be done on any operation, regardless of crop type or geographic region. Key considerations of relevance that may determine measurement approach include whether the farm is irrigated or rain-fed as well as presence of water bodies that are adjacent to the farm. While measurement of the water-use efficiency on the farm may in some cases include consideration of green water such as rain, measurement of water productivity is typically limited to measurement of blue water withdrawals for irrigated crops, and not calculated for rain-fed systems. Water quality metrics can be applied either to water that is adjacent or near to a farm as well as measurement of runoff. However, measurement of water quality is more typically more feasible if measuring adjacent water bodies due to potential high costs of establishing an edge-of-field monitoring system to measure water runoff.

Accuracy – Key processes are included

Calculation of water quantity and quality should take into account assumptions related to importance of the water source including relative contribution from blue or green water, as well as sources of blue water. Both surface and groundwater should be factored into calculation of applied water.

Measurement of water quality can include multiple parameters based on biological, chemical, or physical characteristics. The approach to measurement should consider which of those characteristics are a priority and be transparent about the reasons for selection particular water quality characteristics that are measured and reported.

Credibility – Based on internationally accepted methodology

The choice of approach for outcome measurement metric and calculation as it relates to water quantity and quality outcomes is relatively standard. The main area related to credibility of water quality measurement can include ensuring that proper procedures for water sampling and measurement instruments are used. For example, applied water can be measured directly (e.g., with flow meters) or estimated.

Transparency – Assumptions and calculations available for scrutiny

Key assumptions for measuring water quantity and quality should be clarified and stated when reporting the measurement, including whether the approach measures blue and/or green water use, as well as which specific quality metrics are incorporated.

Ease of use – Provides accessible guidance and is easy to implement

The type of data needed for calculation of water outcome measurement is usually readily known to farmers, but there can be varying degrees of complexity to the metrics. For tools that provide more specialized metrics, there should be sufficient explanation and guidance to help interpret the output.

Perhaps the most important consideration related to ease of use is implementation of approaches to measure water quality. For example, while edge-of-field monitoring may be the most accurate approach for understanding quality of farm runoff it also requires specialized equipment as compared to taking water samples in associated water bodies.

Data Input – Has a simple data entry system

The type of data needed for determination of water metrics is relatively consistent, and most water measurement tools make use of similar data, either relying on information on total applied water for the year or for more details on amount irrigated per week. For water metrics that require data on evapotranspiration (ET) or rainfall, calculator tools may include integrations with sources that provide real-time data updates from weather stations.

Cost – Low and of perceived value

The largest contributor to cost for water outcomes measurement is typically related to associated technologies of measurement on the farm, such as installation of flow meters to measure amount of applied water or cost for water sampling and laboratory tests to determine relevant biological, chemical, or physical properties. For all metrics there is a trade-off between accuracy of the measurement and resources (money and time) that would be needed by the farm to

invest. The farm or farm management group may need to decide on what level of accuracy is reasonable for a given measurement considering the potential costs involved in taking water samples or using the appropriate measurement device.

7.3.1 Additional considerations when choosing a tool or measurement approach

An important additional consideration for measurement of outcomes related to water is the scale of measurement approach and whether outcomes are measured at the level of the farm or the broader watershed. Most available calculation tools for agricultural water outcome measurement focus on the farm level, but some outcome metrics, such as water balance for a catchment area or watershed, may require data beyond the borders of an individual farm or farm group. Landscape level measurement and management of water resources is often necessary when there are multiple farms and other water users in the same area that share the same surface water and groundwater resources and may be organized into irrigation districts. A landscape level approach can take into account water losses in the system before the water reaches the farm (e.g., evaporation that occurs in irrigation canals used to transport water to farms). Determination of broader landscape metrics, such as water balance of the watershed, are helpful to understand broader pressures on water resources but require engagement from multiple stakeholders.

7.4 Example calculators for water metrics

Table 7.2 below provides a summary of examples of available calculators and approach to water measurement in each. General information on some of the tools is provided in a separate section. Because measurement of water quality typically relies only on results from laboratory tests, calculators are generally not required for determination of water quality outcome metrics.

Table 7.2. Calculators available for water outcome measurement

Tool Name	Relevant Metric(s)	Notes
Cool Farm Tool https://cool-farmtool.org/	• Water Productivity (kg/m³) • Total Water Footprint (liter/kg) • Irrigation Efficiency (liter/liter) • Blue/green water footprint (liter/kg) • Water Balance (mm)	
Fieldprint Platform https://calculator.fieldtomarket.org/	• Irrigation Water Use (acre inches per crop yield) • Water quality metric based on Stewardship Tool for Environmental Performance (STEP) developed by US Natural Resources Conservation Service (NRCS) represents potential for nutrient loss from a farm field through runoff or leaching.	Assessment of water quality within the Fieldprint Platform is a proxy for outcome measures and uses NRCS Conservation Effects Assessment Project (CEAP) model to evaluate risk of nutrient loss based on land management and conservation practices for a given soil and climate.
Stewardship Index for Specialty Crops (SISC) https://www.stewardshipindex.org/	• Simple Irrigation Efficiency (unit of water applied divided by unit of water demand; %) • Applied Water Use Efficiency (water volume applied divided by crop weight)	Calculation approach for irrigation efficiency is published on the SISC website with instructions on calculation approach for growers.

8 Example calculator tools

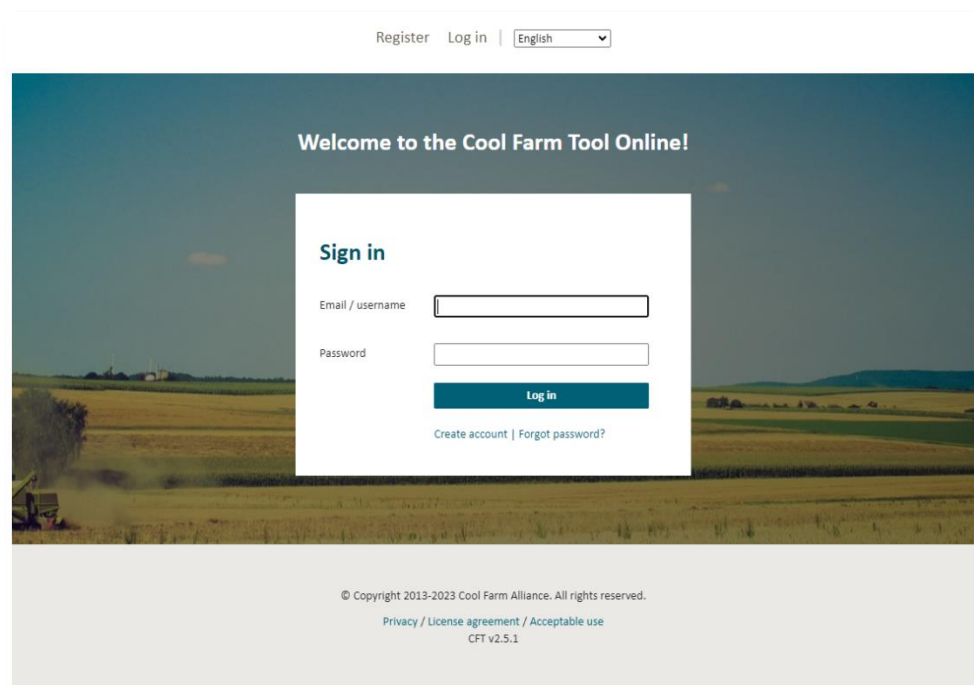
Three calculator tools are described here to provide FSA users with additional context and example tools that could be used for their supply chain. These are examples to consider for FSA users and additional tools may be added in the future. There are several other tools that are available (e.g., Columb et al., 2013; Deneff et al., 2012; GrainGrowers & Kondinin, 2020; WRI and WBCSD, 2022 - Table III) and users are encouraged to review options according to key considerations as described in this Handbook to determine the right tool for their supply chain or FMG. In some cases, the best tool may be focused on a particular outcome measurement, region, or crop system.

8.1.1 Cool Farm Tool

<https://app.coolfarmtool.org/>

Usability Considerations	Observations
Outcome Metrics	GHGs/climate; water; biodiversity
Cost/Relevance	Free for up to 5 evaluations/Global. Annual, perennial, and livestock available. Covers farm-level activities to farm gate.
Support	info@coolfarmtool.org. Onscreen and PDF help available.
Time Investment	30-45 minutes per field. Copy function can cut down time if practices are similar between fields.
Learning Opportunity	Graphical display of contribution of total emissions by factor can show farmers where GHGs are high vs. low for each field. Comparing two fields side by side is a powerful reporting option.
Additional Comments	Simplified crop protection product and fertilizer product entry. Data is entered by field or by crop which requires farmers with multiple crops to allocate inputs and energy use appropriately.

Cool Farm Tool data entry is divided into seven tabs, Crop, Soil, Inputs, Fuel and Energy, Irrigation, Carbon, and Transport. The Crop and Soil tabs have a defined set of variables to enter whereas the other tabs are open-ended to accommodate a varying number of activities.

The image shows the login page for the Cool Farm Tool Online. At the top, there are links for 'Register' and 'Log in', and a language dropdown menu set to 'English'. The main heading is 'Welcome to the Cool Farm Tool Online!'. Below this is a 'Sign in' section with two input fields: 'Email / username' and 'Password'. A blue 'Log in' button is positioned below the password field. Underneath the button are links for 'Create account' and 'Forgot password?'. The background of the page is a scenic image of a farm with rolling green hills and a blue sky. At the bottom, there is a copyright notice: '© Copyright 2013-2023 Cool Farm Alliance. All rights reserved.' and links for 'Privacy / License agreement / Acceptable use' with the version 'CFT v2.5.1'.

On the Inputs tab, fertilizer inputs can be aggregated by fertilizer product and entered as a sum for the season which can streamline entry. Crop protection inputs are divided into three types, seed treatments, soil treatments and post emergent treatments. Seed and soil treatments must be entered as distinct applications, but post-emergent applications may be entered as a single value across the growing season.

Fuel and Energy can be entered in aggregate by fuel type and include the option to label each entry. Alternatively, there are energy-use estimators to estimate fuel used by machines and operation. There is a similar treatment for energy requirements for irrigation, meaning users can enter a total energy use or use an estimator based on irrigation details.

The transport tab estimates energy used to move products to or from the field such as transportation of manure to the field or finished product to a processor or storage facility.

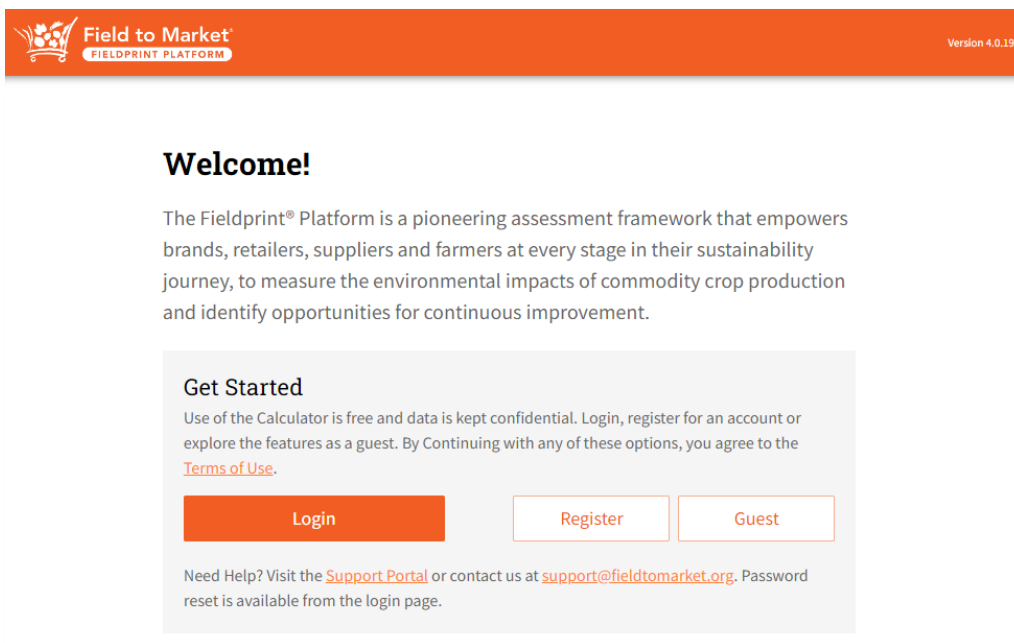
Some results appear as a sidebar in real time as the user enters and saves inputs. Full results are presented in the following tabs, GHG, Compare, Performance, Costs, and Data and shown as total amount, per acre and per production unit as both a graph and table. Actions to change results are not suggested within the tool but graphs show key contributors. Users can copy data from one field to another field which may more easily allow comparison of results from changing practices.

8.1.2 Fieldprint Platform

<https://calculator.fieldtomarket.org>

Usability Considerations	Observations
Outcome Metrics	GHGs/climate; water; biodiversity; soil
Cost/Relevance	Free/US and Canada. Annual crops. Covers farm-level activities to farm gate.
Support	support@fieldtomarket.org . Onscreen and PDF help available.
Time Investment	Once crop rotations are established, about 40 minutes per field. Copy function and rotation templates can reduce time if practices are similar between fields.
Learning Opportunity	Results presented as spidergram graphic or excel output. US state and national benchmarks are available where applicable.
Additional Comments	Comprehensive sustainability tool providing metrics across eight sustainability areas (Biodiversity, Energy Use, Greenhouse Gas Emissions, Irrigation Water Use, Land Use Efficiency, Soil Carbon, Soil Conservation, and Water Quality).

Fieldprint Platform data is entered in a multistep process: (1) Definition and location of fields; (2) Rotation and Residue practices; (3) Management Information; (4) Operations which are broken down to (4a) planting, (4b) Irrigation and (4c) Harvesting and (4d) Fertilizer and Spraying operations.



Defining the crop rotation and residue practices is required so the tool can use models for soil erosion and nitrous oxide (N₂O) emissions based on crop history and location. A search of the rotation library found 7,218 rotations containing *corn* and 3,104 rotations containing *soy*. Filters by geography or keyword can help choose an accurate rotation. Alternatively, users can define their own rotations, including implement passes, fertilizations, and harvest events for up to 5 years. The section labeled Management Information, requests information on a defined set of management practices and decision making. The section on Operations Information, includes planting, irrigation (which can be entered directly or by estimator) and harvesting all of which have a set number of inputs. The Fertilizer and Spraying Operations section accommodates as many trips as needed for the application of fertilizers, manure, or crop protectants.

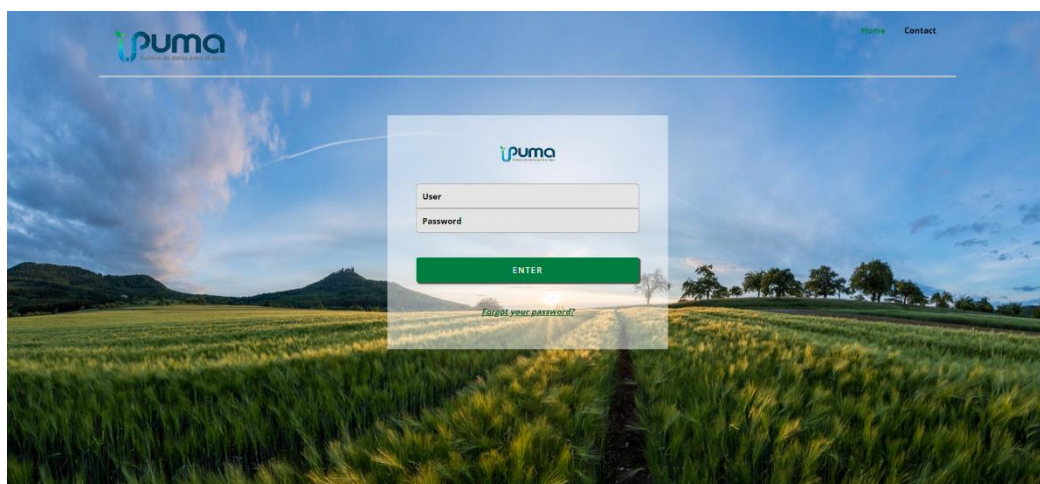
Results are displayed after sending data for Fieldprint Analysis which can take several seconds or more. The Fieldprint Analysis returns a spidergram of eight metrics of the sustainability for commodity crop production - Biodiversity, Energy Use, Greenhouse Gas Emissions, Irrigation Water Use, Land Use Efficiency, Soil Carbon, Soil Conservation, and Water Quality. Spidergram results show normalized data on a 0-to-100-point scale where smaller values indicate less resource use or environmental impact. Detailed reports are available as PDF or Excel export, which offer farmers the ability to analyze components factoring into their sustainability outcome, evaluated by acre and per unit of production.

8.1.3 Puma Platform

<https://www.plataformapuma.com/>

Usability Considerations	Observations
Outcome Metrics	GHGs/climate; soil
Cost/Relevance	Paid access, annual subscription on a per ha basis. Country specific (Argentina, Brazil, Uruguay) but easy worldwide deployment potential. Annual and perennial crops. Covers farm-level activities to farm gate.
Support	info@plataformapuma.com and customer service support. User manual and tutorials available.
Time Investment	Crop carbon footprint takes 30-45 minutes per field. Possible to import data from Excel and accounting software. Multi-year simulation tool takes 1-1.5 hours (copy function for similar agronomic practices/crops).
Learning Opportunity	Crop carbon footprint data, presented on a per year basis or multi-year simulation, give farmers a powerful tool to learn about improving agronomic practices for enhancing productivity and sustainability.
Additional Comments	Aggregation of results by crop/farm/region/country is possible. Crop carbon footprint data can be exported as tables and graphs in PDF/Excel/API format. Future updates of the tool also include forestry, rice, sugar cane and live-stock production.

Puma is an environmental monitoring platform which supports users to assess crop production cycle impact on carbon footprint and soil carbon balance, providing sustainability reports in line with international guidelines (e.g., Intergovernmental Panel on Climate Change - IPCC, Biomass Biofuel Sustainability Voluntary Scheme – 2BSvs).



Data entry steps include: (1) Georeferenced field boundaries, and (2) Crop agronomic management data is entered either online, from Excel, or from linking to the farmer's accounting system.

- Crop tab: user can select the required international certification guideline, cropping system, planted and harvested acreage, grain yield and crop residue removal.
- Tillage tab: tillage, fertilizer, seed, and crop protection products are detailed using country-specific agrochemical data and emissions factors.
- Irrigation tab: allows user to select irrigation system, events (date and volume) and type of energy source.
- Incidents tab: captures weather events that may affect crop production and residue (i.e., frost, flood, fire).
- Soil tab: allows the user to enter soil organic carbon, clay percentage and bulk density.

The tool uses either georeferenced soil and climate data as standard (from open-sourced models) or farm-specific soil data (e.g., soil tests). Based on on-farm activity and grain yield, soil organic carbon and crop biomass are estimated and used to present values for soil carbon balance and organic matter variation.

Carbon footprint results are divided into sources of emissions, with their corresponding greenhouse gases, expressed by hectare and ton. Reports can be downloaded as a PDF. Soil carbon balance information is also presented (equilibrium biomass for C neutrality, gain/loss of both organic matter and soil organic carbon). The output tables and graphs illustrate emissions hotspots and areas for improvement.

The simulation model tool can be used to compare alternative crop production scenarios (e.g., crop rotation, tillage, fertilizer management), for a desired number of cropping seasons, to understand their impact on carbon footprint and soil carbon balance on a single crop or the assessed cropping seasons.

9 Glossary of Terms

AGRO-BIODIVERSITY	The variety and variability of animals, plants and micro-organisms that are used directly or indirectly for food and agriculture, including crops, livestock, forestry, and fisheries. It comprises the diversity of genetic resources and species used for food, fodder, fiber, fuel, and pharmaceuticals. It also includes the diversity of non-harvested species that support production, and those in the wider environment that support agro-ecosystems and their diversity.
BIODIVERSITY	The variability among living organisms from all sources and the ecological complexes of which they are a part; this includes diversity within species, between species and of ecosystems.
BLUE WATER	Fresh surface and groundwater. In other words, the water in freshwater lakes, rivers, and aquifers (Water Footprint Network, 2023)
CALCULATOR	An automated web-, excel-, or other software-based calculation tool, developed for quantifying outcomes such as GHG emissions, water, soil health, biodiversity, or other features as they relate to farm operations (<i>modified from</i> Denef et al., 2012).
ECOSYSTEM	A dynamic complex of plant, animal and microorganism communities and their non-living environment interacting as a functional unit.
ECOSYSTEM SERVICES	Any direct or indirect benefit that nature provides to society.
EVAPOTRANSPIRATION	The combined process of water leaving the surface of the earth to the atmosphere, including evaporation from the soil and transpiration from plants.

FARM MANAGEMENT GROUP (FMG)	A group of farms with a logical coherence asserting that it meets the requirements of the FSA implementation framework as a group.
FERTILIZER	A material applied to the soil or crop to introduce one or more essential nutrient for plant growth.
FOOTPRINT	Total amount of a resource used (e.g., water) or emitted (e.g., carbon) by a person, operation, or organization.
GREEN WATER	The precipitation on land that does not run off or recharge the groundwater but is stored in the soil or temporarily stays on top of the soil or vegetation. Eventually, this part of precipitation evaporates or transpires through plants. Green water can be made productive for crop growth (although not all green water can be taken up by crops, because there will always be evaporation from the soil and because not all periods of the year or areas are suitable for crop growth) (Water Footprint Network, 2023).
GROUND WATER	Water below the surface of the Earth stored in pore spaces and fractures within rock or layers of sand and gravel (aquifers) (AWS, 2019).
HABITAT	A place or type of site where a species population naturally occurs
IRRIGATION EFFICIENCY	Water applied divided by total water demand of the crop.
METRIC	A system or standard of measurement (from Jennings et al. 2020).
NUTRIENTS	These are essential elements required by the crop to survive, grow, and reach full potential. <i>Macronutrients:</i> This includes Nitrogen (N), Phosphorus (P), Potassium (K) and Sulfur (S).

Micronutrients: This includes Boron (B), Calcium (Ca), Chlorine (Cl), Copper (Cu), Iron (Fe), Magnesium (Mg), Manganese (Mn), Molybdenum (Mo), and Zinc (Zn).

OUTCOME	The short-term and medium-term effects of an intervention on the sustainability issue in question (from Jennings et al. 2020).
REHABILITATION	Management actions that aim to reinstate a level of ecosystem functioning on degraded sites, where the goal is renewed and ongoing provision of ecosystem services rather than the biodiversity and integrity of a designated native reference ecosystem (<i>from</i> Gerwing et al. 2021).
RESILIENCE	Ability to withstand and recover from biophysical or socio-political shocks.
ROOT ZONE	The area of soil and pores surrounding the root structure.
SOIL	The dynamic interface between the lithosphere (rock), atmosphere (air), hydrosphere (water), and biosphere (living things).
SOIL AMENDMENTS	Minerals used to improve soil fertility. Examples include gypsum for improved aeration and lime and sulfur for pH optimization. Certain essential nutrients may be applied to optimize the base ratio. Organic amendments are covered in Soil Organic Matter.
SOIL COMPACTION	Soil compaction is a common term related to soil density and can be measured by a penetrometer. As a soil becomes more compact, its pore space is reduced, allowing less room for water and air. This is indicated by a greater bulk density and reduces available water and air. The risk of soil compaction is greater with heavy operations on high clay content and also high moisture content.

SOIL EROSION	The displacement of topsoil, resulting in a loss of fertility. Airborne (wind erosion) occurs when topsoil is displaced by wind. Airborne erosion is accelerated with poor ground cover, overworked soil, and lack of windbreaks. Waterborne erosion occurs when topsoil is displaced by water and is accelerated with poor soil aggregation, poor surface drainage, poor ground cover and steep slope (gradient) of land.
SOIL HEALTH	Soil health is the capacity of soil to function as a living system, with ecosystem and land use boundaries, to sustain plant and animal productivity, maintain or enhance water and air quality, and promote plant and animal health. Healthy soils maintain a diverse community of soil organisms that help to control plant disease, insect and weed pests, form beneficial symbiotic associations with plant roots; recycle essential plant nutrients; improve soil structure with positive repercussions for soil water and nutrient holding capacity, and ultimately improve crop production. A healthy soil does not pollute its environment and does contribute to mitigating climate change by maintaining or increasing its carbon content. Soil health can be defined as a three-pillared management of the physical, chemical, and biological soil aspects.
SOIL MOISTURE	The amount of water contained in the soil pores. Saturated soils are where the pores are 100% occupied by water, leaving no space for air in the crop root zone. Soil moisture content is expressed as a percentage of water mass over dry mass of the soil.
SOIL ORGANIC CARBON (SOC)	SOC is used to estimate organic matter levels. Organic matter is about 60% carbon.
SOIL ORGANIC MATTER (SOM)	The SOM refers to organic material that is living, or once living, and applies to organic matter present in or added to the soil in the form of mulch or compost.

SOIL PROFILE, SOIL HORIZON	The soil horizon is a layer parallel to the soil surface whose physical, chemical, and biological characteristics differ from the layers above and beneath.
SOIL RESPIRATION	The production of carbon dioxide when soil organisms respire. Also called, Carbon Mineralization Potential (burst of CO ₂), this is a lab test that measures the biological activity of soils.
SOIL, TOPSOIL	The upper, outermost layer of soil with the highest concentration of organic matter and microorganisms and is where most of the Earth's biological soil activity occurs.
SOIL WET AGGREGATE STABILITY	Wet aggregate stability is a measure of a dry soil aggregate's ability to resist dispersion once rewetted. The unit is % WSA weight of stable aggregates/ total weight.
SPECIES	A group of organisms with common attributes, typically defined as a group that can reproduce with one another and produce fertile offspring.
SPECIES EVENNESS	The distribution of abundance across species in a community. Species evenness is highest when all species in a community have similar number of individuals and is lower if one species is dominant and all other species have very few individuals (Pyron, 2010). See also <i>Species Richness</i>
SPECIES RICHNESS	The number of species within a defined community or region (Pyron, 2010). See also <i>Species Evenness</i>
SURFACE WATER	Water located on the surface of the Earth, such as in streams, rivers, and lakes (IGRAC, 2023).

WATER FOOTPRINT	Volume of water consumption per unit of crop. Green water footprint is a measure of rainwater consumed; blue water footprint measure amount of surface and ground water consumed (<i>modified from Mekonnen and Hoekstra, 2014</i>).
WATER BALANCE	An assessment of all water flows and storage volumes of an entity. Water balance can apply to a specific farm site or separately for the catchment. The assessment should measure all water inflows, throughflows, outflows, water storage volume and changes in storage. The first step is to identify and map each component, and then to quantify it. These are combined into the water balance equation, which should balance (at least approximately): {water outflow} = {water inflow} + {change in storage}. Sustainable water balance is the condition whereby ongoing water use in the catchment has no long-term negative impact on the natural environment and legitimate water users (<i>modified from AWS, 2019</i>).
WATER PRODUCTIVITY	Usually defined as crop yield or production per unit of water (e.g., kg of crop per m ³ of water).
WATERSHED	An area of land that channels rainfall and snowmelt to creeks, streams, and rivers, and eventually to outflow points such as reservoirs, bays, and the ocean. Typically used interchangeably with the terms “drainage basin” or “water catchment” (<i>modified from NOAA, 2023</i>).

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